UNIT 1 · LESSON 1

Introduction to Waves

What a wave is, the two kinds of waves, and how we measure them.

In this lesson you will learn...

- What a wave is (and what it isn't)
- Transverse vs. longitudinal waves
- Amplitude, wavelength and frequency
- How to calculate wave speed

Words to know

Wave
A disturbance that carries energy from one place to another.

Medium
The material a wave travels through: solid, liquid or gas.

Amplitude
How far the medium moves away from its resting position.

Wavelength
The length of one full wave, e.g. crest to crest.

Frequency
How many waves pass a point each second. Unit: hertz (Hz).

Wave speed
How fast the wave travels through the medium.

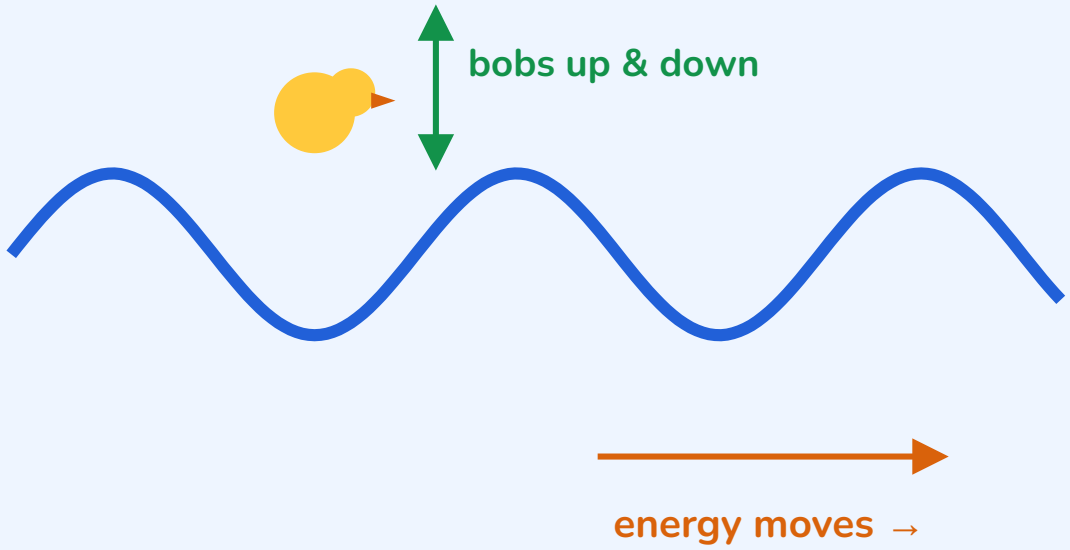
Think about it

At a stadium, the crowd does “the wave.” The wave goes all the way around the stadium.

Did anyone in the crowd actually travel around the stadium?

A wave moves energy, not matter

- A wave is a **disturbance** that carries **energy** from one place to another.
- The material (the **medium**) only wiggles in place. It does not travel with the wave.
- Bigger disturbance → more energy carried.
- Stadium answer: each person just stands up and sits down. Only the energy goes around!



The duck stays put. Only energy travels.

Pulse or wave?

- One single push makes a **pulse**: one bump of energy.
- Repeated pushes make a **wave**: a pattern that repeats.
- Example: flick a rope once → pulse. Keep shaking it → wave.

Pulse: one disturbance



Wave: repeating disturbances



Two families of waves

Mechanical waves

Need a medium to travel.

Examples: sound, water waves, earthquake (seismic) waves.

They cannot travel through empty space.

Electromagnetic (EM) waves

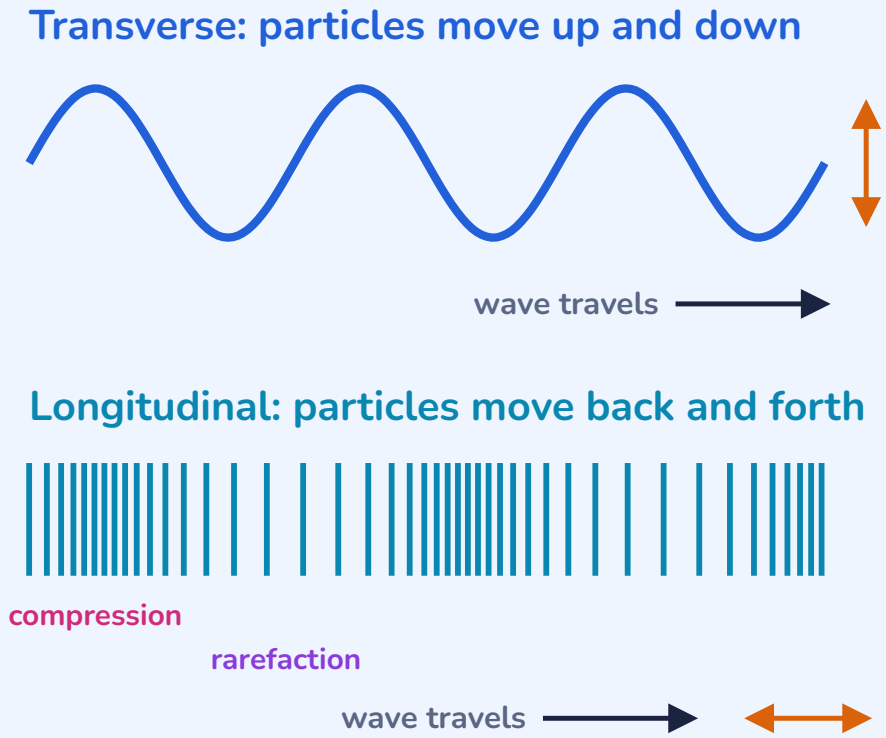
Do not need a medium. They can cross empty space.

Examples: light, radio waves, microwaves, X-rays.

This is how sunlight reaches Earth.

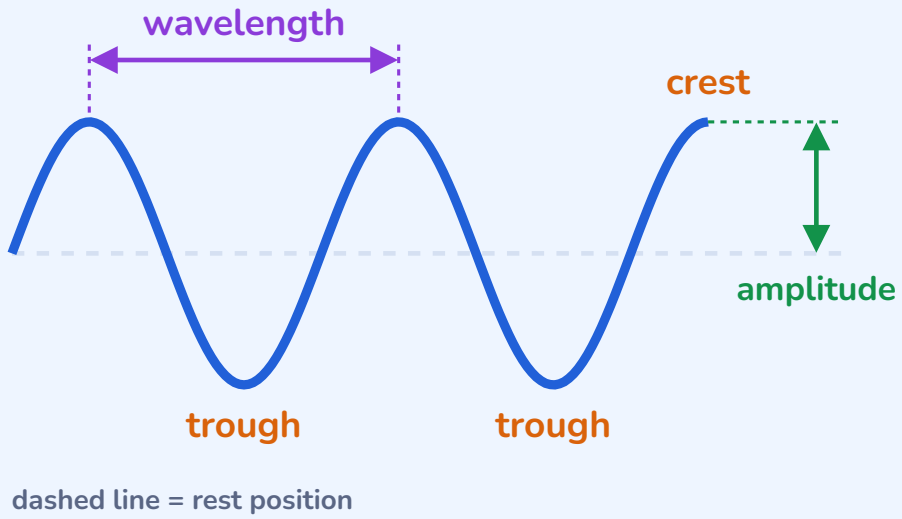
Transverse vs. longitudinal

- **Transverse:** the medium moves at a right angle (⊥) to the direction the wave travels. Rope waves, light, earthquake **S waves**.
- **Longitudinal** (compressional): the medium moves back and forth in the **same** direction the wave travels. Sound, earthquake **P waves**.



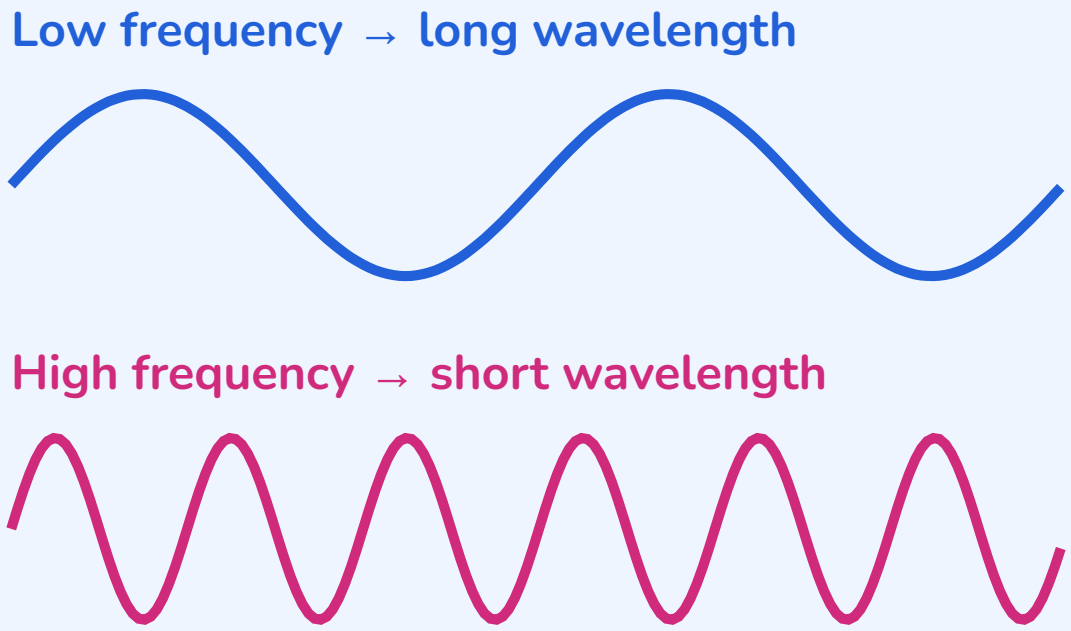
Parts of a wave

- **Crest:** highest point. **Trough:** lowest point.
- **Amplitude:** from the rest line up to a crest (or down to a trough).
- **Wavelength (λ):** crest to crest, or trough to trough. Measured in meters.
- Longitudinal waves use **compressions** (squeezed) and **rarefactions** (spread out) instead.



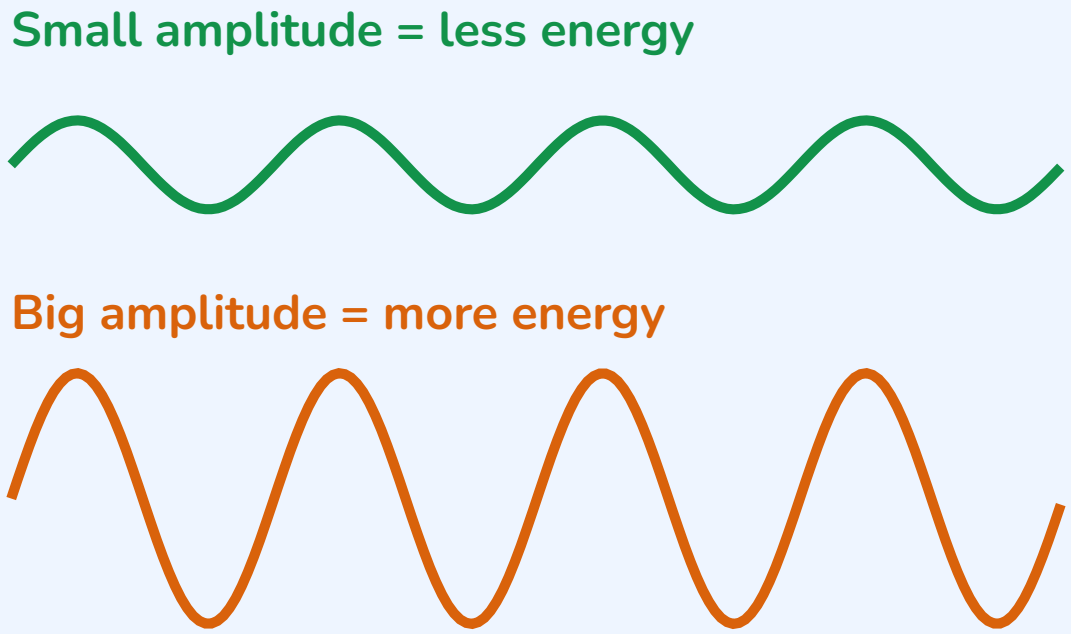
Frequency

- **Frequency** = how many waves pass a point in one second.
- Unit: **hertz (Hz)**. 1 Hz = one wave per second.
- In the same medium: **higher frequency → shorter wavelength**.
- Count crests (transverse) or compressions (longitudinal) passing by.



Amplitude and energy

- More energy → bigger amplitude. Pluck a guitar string harder and it moves farther.
- Energy grows with amplitude **squared**.
- Amplitude × 2 → energy × 4.
  Amplitude × 3 → energy × 9.
- Working backwards: energy × 16 → amplitude × 4 (because 4 × 4 = 16).



Wave speed

speed = wavelength × frequency · $v = \lambda \times f$

- Example: λ = 2 m, f = 3 Hz
  v = 2 × 3 = **6 m/s**
- Speed depends on the **medium**.
- Sound: faster in solids and liquids than in air, and faster in warm air.
- Light: fastest in empty space, slower in water or glass.

💡 Your turn: a wave has λ = 0.5 m and f = 10 Hz. How fast is it? (Answer on the last slide.)

Check yourself

1. A cork floats on a pond as a wave passes. Where is the cork after the wave goes by? Why?
2. Sound can't travel through outer space, but light can. Use the words **mechanical** and **electromagnetic** to explain.
3. Wave A has twice the amplitude of wave B. How many times more energy does wave A carry?
4. A wave has a wavelength of 4 m and a frequency of 5 Hz. What is its speed?
5. Is sound a transverse or a longitudinal wave? How do you know?

Answers

1. About where it started. The wave carries energy, not the water or the cork.
2. Sound is mechanical, so it needs a medium. Light is electromagnetic, so it can cross empty space.
3. 4 times (2 squared = 4).
4. v = 4 × 5 = 20 m/s. (Your turn slide: 0.5 × 10 = 5 m/s.)
5. Longitudinal. The air moves back and forth in the same direction the sound travels.